

Donggyu Joo

SENIOR MACHINE LEARNING SCIENTIST @ SAMSUNG RESEARCH

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O-1 Visa Eligible (No H-1B Lottery Required): 2,800+ Citations & 5+ Top-tier Publications

Summary

Senior Machine Learning Scientist with a **Ph.D. in AI** and 3+ years at **Samsung Research**, building on-device computer vision and vision-language AI for Samsung Galaxy devices. Shipped **8 on-device AI features** globally, including VLM-based semantic photo search for Galaxy Gallery with models under 100M parameters. Expertise spans large-scale multimodal data curation (50M+ image-text pairs), model training and evaluation, on-device optimization, and robustness improvement for real-world edge cases. **Team Lead** for 6 engineers, driving cross-functional execution across research, product, and engineering.

Skills

Core AI & Research	Computer Vision, Vision-Language Models, Image-Text Retrieval, Generative Modeling, Model Compression
Production ML	On-device AI, Model Optimization, Quantization, Distillation, TFLite Deployment, Edge-case Debugging
Data & Evaluation	Large-scale Data Curation, Model Training & Evaluation, Robustness Evaluation, Data Quality Control
Languages & Tools	Python, PyTorch, TensorFlow, TFLite, Git

Experience

Samsung Research, Samsung Electronics

Seoul, S. Korea

Senior Machine Learning Scientist

Sep. 2022 - Present

- **On-Device VLM & Gallery Search:** Developed and deployed VLM-based text-to-image retrieval models for Samsung Galaxy Gallery, enabling semantic photo search with models under 100M parameters. Expanded supported query coverage from 500 to 2,000+ categories, including documents, landmarks, and car brands.
- **Multimodal Data Curation:** Built GenAI-assisted labeling and filtering pipelines for 50M+ image-text pairs. Improved robustness by reducing noisy labels and increasing coverage of long-tail real-world image categories.
- **Research-to-Product Deployment:** Shipped 8 on-device AI models to production for Galaxy devices. Owned model optimization, TFLite conversion, on-device validation, latency/memory tuning, and edge-case debugging based on user feedback.
- **Touch Sequence Classification:** Designed and optimized an on-device sequential classification model to detect unintended pocket touches. Achieved sensor-level reliability under challenging edge cases and deployed real-time CPU inference to Galaxy Z Flip6/7.
- **Technical Leadership:** Led 6 engineers across data, modeling, evaluation, and release. Collaborated with product and engineering stakeholders to turn ambiguous user-facing problems into deployable AI features.

Education

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)

Daejeon, S. Korea

Ph.D. in Electrical Engineering

Mar. 2018 - Aug. 2022

- Research Focus: Model Compression, Model Pruning, and Computer Vision

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)

Daejeon, S. Korea

M.S. in Electrical Engineering

Sep. 2016 - Feb. 2018

- Research Focus: Generative Adversarial Networks (GANs) and Computer Vision

KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST)

Daejeon, S. Korea

B.S. in Electrical Engineering (Summa Cum Laude)

Feb. 2013 - Aug. 2016

Selected Publications

- **Donggyu Joo**, Eojindl Yi, Sunghyun Baek, and Junmo Kim, “Linearly Replaceable Filters for Deep Network Channel Pruning”, The Thirty-Fifth AAAI Conference on Artificial Intelligence (**AAAI**), 2021
- Janghyun Lee, Hyeonngwon Hong, **Donggyu Joo**, and Junmo Kim, “Continual Learning with Extended Kronecker-factored Approximate Curvature”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2020
- Janghyun Lee, **Donggyu Joo**, Hyeonngwon Hong, and Junmo Kim, “Residual Continual Learning”, Thirty-Fourth AAAI Conference on Artificial Intelligence (**AAAI**), Oral presentation, 2020
- **Donggyu Joo***, Doyeon Kim*, and Junmo Kim, “Generating a Fusion Image: One’s Identity and Another’s Shape”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2018
- Junho Yim, **Donggyu Joo**, and Junmo Kim, “A Gift From Knowledge Distillation: Fast Optimization, Network Minimization and Transfer Learning”, IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2017

Other Publications

- Doyeon Kim, Dongyeun Lee, **Donggyu Joo**, Junmo Kim, “Training Cartoonization Network without Cartoon”, 2023 IEEE International Conference on Image Processing (**ICIP**), 2023
- **Donggyu Joo**, Sunghun Baek, and Junmo Kim, “Which Metrics For Network Pruning: Final Accuracy? or Accuracy Drop?”, The 29th IEEE International Conference on Image Processing (**ICIP**), 2022
- Doyeon Kim, Woonghyun Ga, Pyungwhan Ahn, **Donggyu Joo**, Sehwan Chun, Junmo Kim, “Global-Local Path Networks for Monocular Depth Estimation with Vertical CutDepth”, arXiv preprint arXiv:2201.07436, 2022
- **Donggyu Joo**, Doyeon Kim, Eojindl Yi, and Junmo Kim, “Linear Combination Approximation of Feature for Channel Pruning”, Efficient Deep Learning for Computer Vision CVPR Workshop (**CVPRW**), 2022
- **Donggyu Joo**, Doyeon Kim, and Junmo Kim, “Slimming Resnet by Slimming Shortcut”, 25th International Conference on Pattern Recognition (**ICPR**), 2020.
- Doyeon Kim, **Donggyu Joo**, and Junmo Kim, “Delivering Meaningful Representation for Monocular Depth Estimation”, 25th International Conference on Pattern Recognition (**ICPR**), 2020.
- Doyeon Kim*, **Donggyu Joo***, and Junmo Kim, “TiVGAN: Text to Image to Video Generation with Step-by-Step Evolutionary Generator”, **IEEE Access**, 2020.
- **Donggyu Joo**, Junho Yim, and Junmo Kim, “Unconstrained Control of Feature Map Size Using Non-integer Strided Sampling”, 29th British Machine Vision Conference (**BMVC**), 2018
- **Donggyu Joo**, Youngdong Kim, and Junmo Kim, “Identity Preserving Residual-Spatial Transformer Network”, The 5th International Conference on Robot Intelligence Technology and Applications (**RiTA**), Oral presentation, 2017

Honors & Awards

- 2020 **Scholarship**, Young-Han Kim Global Leader Scholarship (\$5,000)
- 2019 **Winner**, AI Reporter, AI WorldCup (\$5,000)
- 2018 **Winner**, AI Reporter, AI WorldCup (\$5,000)
- 2018 **Silver Prize**, 24th Samsung Humantech Paper Awards, Signal Processing (\$7,000)
- 2017 **Silver Prize**, 23rd Samsung Humantech Paper Awards, Computer Science (\$7,000)
- 2016 **Summa Cum Laude**, School of Electrical Engineering, KAIST
- 2015 **Dean’s List**, School of Electrical Engineering, KAIST

Neural Network Learning Method and Device for Recognizing Class

Junmo Kim, Byungju Kim, Juchang Kim, Yeakang Lee, Minsuk Park, Juseung Yun, Jaeyoung Lee, **Donggyu Joo**
US 11,449,733 B2, U.S. Patent (Registered)

Electronic Device for Processing Image and Operating Method Thereof

Donggyu Joo, Dongchan Kim, Jinyoung Hwang
US 2025/0292376 A1, U.S. Patent Application

ELECTRONIC DEVICE AND OPERATING METHOD FOR THE SAME

Sangwon Lee, Sangmin Yoo, **Donggyu Joo**, Jiman Kim, Jinyoung Hwang
10-2024-0114486, Republic of Korea Patent Application

Knowledge Transfer Method Using Deep Neural Network and Apparatus Therefor

Junmo Kim, Junho Yim, **Donggyu Joo**
10-2061935, Republic of Korea Patent (Registered)

Incremental Training Based Knowledge Transfer Method for Training Large Deep Neural Networks and Apparatus Therefor

Junmo Kim, Junho Yim, **Donggyu Joo**, Jihoon Bae, Naesoo Kim, Chulsik Pyo
10-2063377, Republic of Korea Patent (Registered)

Knowledge Sharing Based Knowledge Transfer Method for Improving Quality of Knowledge and Apparatus Therefor

Junmo Kim, Junho Yim, **Donggyu Joo**, Jihoon Bae, Naesoo Kim, Chulsik Pyo
10-2063376, Republic of Korea Patent (Registered)

Electronic Apparatus for Identifying Abnormal Touch and Control Method Thereof

Donggyu Joo, Jinyoung Hwang
10-2024-0009651, Republic of Korea Patent Application